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Passive acoustic identification of bubble flow regime based on synchrosqueezing wavelet transform and deep learning

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Abstract

A passive acoustic method based on synchrosqueezing wavelet transform (SWT) and deep learning was proposed to automatically identify bubble flow regimes in process industries. The method was established on the bijection relationship between bubble flow regime and the time-frequency (TF) textures of its passive acoustic emission (PAE) signals. Specifically, the PAE signal of the bubble flow was first acquired by hydrophones, then converted into TF representations (TFRs) by SWT and finally used to train convolutional neural network (CNN) to identify bubble flow regimes automatically. The effects of TF analysis method and CNN architecture were studied: the SWT represented TF textures of PAE signals the best, and its combination with ResNet gave the highest identification accuracy of 99.67%. As its main contribution, the method validated the feasibility of PAE and CNN in bubble flow regime identification, and achieved automatic identification of bubble flow regime with high accuracy and barely no subjective interference.

KEYWORDS
bubble flow, convolutional neural network, flow regime identification, passive acoustic emission, synchrosqueezing wavelet transform

1 | INTRODUCTION

Bubble flow is a special kind of gas–liquid two-phase flow in which the gas scatters in the fluid as bubbles, commonly seen in many industrial processes, such as chemical engineering, petroleum production, and power generation.[1] The size and space distribution of bubbles in the liquid, generally known as bubble flow regime, is quite complex, and is also one of the crucial parameters influencing the production efficiency in those processes. Thus it is necessary but also challenging to identify bubble flow regime in time and keep it under best condition so as to maximize the benefits.[2,3] There have been lots of methods for bubble flow regime identification, generally classified into subjective and objective methods considering whether it is dominated by human judgments. The subjective methods are currently the main way to identify bubble flow regimes,[4] which make identifications by human judgments based on visual observations/cameras and experiences. It is easy, cheap, and convenient, but it has strict requirements on observers and environment, such as rich experiences for observers and transparent vessels, which is hard to meet in real industrial processes, and even if met the results may still vary from person to person. Therefore, it is unreliable and unfeasible for the subjective methods to be applied in industries.

To make up for the shortcomings of subjective methods, lots of objective methods have been developed to better identify bubble flow regimes so as to meet industrial needs. The objective methods generally have four steps: signal acquisition, feature extraction, classifier definition, and regime identification, and can be of many kinds depending on the signal types used, such as differential pressure methods,[5-7] electrical properties methods (conductivity, resistivity, capacity),[8-10] optical properties methods,[11] acoustic properties methods,[1,4,12] high speed imaging methods,[13,14] and radiation

Abbreviations: CNN, convolutional neural network; CWT, continuous wavelet transform; DL, deep learning; FFT, fast Fourier transform; PAE, passive acoustic emission; SNR, signal to noise ratio; STFT, short time Fourier transform; SWT, synchrosqueezing wavelet transform; TF, time-frequency; TFR, time-frequency representation; WT, wavelet transform; VAR, variance.

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attenuation methods.[15-17] Among these methods, acoustic methods are gaining more and more attention lately. The acoustic methods utilize sound signals to distinguish bubble flow regimes, and there are two kinds of them considering the source of sound signal, namely positive and passive methods. The word "positive" means that the sound is intentionally generated by human, and the intensity or time-frequency (TF) change of the sound after going through or being reflected by the mixture is what is used to identify flow regimes. The sound used in positive method is often ultrasonic,[1,4] which, with no doubts, always means a complicated, and expensive experimental system. As for the passive ones, the sound is from bubble flows. The bubble flow emits sound signals during the process of generation, rising, breaking up, and coalescence, and the sound texture of each bubble flow regime is unique thus perfect for telling the regimes apart. Verified by Husin et al.[18] to be feasible in flow regime identification, Pandit et al.[19] determined the bubble size distribution in water by sound emission, furthermore, Fang et al.[20] and Maier et al.[12] have successfully linked the sound characteristics with flow regimes and proved its effectiveness in bubble flow regime identification. Compared with positive methods, the passive ones are easier and cheaper to execute with only several hydrophones needed, and therefore the final choice of this paper.

To get feature extraction, classifier definition and regime identification jobs done properly, deep learning (DL) is now preferred instead of statistical analysis or the combination of the two as a result of its objectivity and accuracy. The convolutional neural network (CNN) is a kind of DL architectures, and it is fast, robust, and highly invariant to translation, scaling, tilting, or other forms of image distortion.[21,22] CNN learns features implicitly and makes identification from the training process at the same time, completely avoiding explicit features extraction or features pre-defining which is necessary in other methods like statistical analysis so as to exclude human interference and gain a higher identification accuracy, thus it has been widely used in fields like image recognition and classification and proved to be quite effective.[23-25] It can be foreseen that CNN can do a good job in spectrogram based bubble flow regimes identification as well.

It is easy to directly use the time series of sound signals as the input of CNN, but it comes along with drawbacks like inaccurate features extraction and lower identification accuracy. Therefore, TF analysis is usually necessary for highlighting signal features for CNN to make better identification, and the synchrosqueezing wavelet transform (SWT) has shown to be a more effective TF analysis method than short time Fourier transform (STFT) or wavelet transform (WT) for acquiring higher TF resolution spectrograms of signals.[26,27] The SWT is a TF domain redistribution method based on WT.[28] It takes advantage of the fact that the phase of signal in frequency domain is not affected by scale transform to get the corresponding frequency of each scale, and then adds up the scales under the same frequency, that is, redistributes and squeezes the wavelet coefficients obtained by WT, so as to squeeze the scale values near certain frequency into the very same frequency to improve the blurring in the scale direction as well as the TF resolution.[26]

In this paper, a passive acoustic bubble flow regime identification method based on SWT and CNN is proposed. The bubble flow is generated in an acrylic cubic container by an annular bubble generator on the bottom. The passive acoustic emission (PAE) signals at three different positions are first acquired by three hydrophones fixed along the axial direction of the bubble flow, and the bubble flow images are recorded by a high-speed camera simultaneously. Then the signals are pre-processed by SWT to get the TF representations (TFRs), upon which a dataset consisting of a training set, a validation set, and a test set can be built. Next, the dataset is labeled with flow regime by the bubble flow images recorded. Finally, the CNN is trained, validated and tested on the dataset to extract the key TF textures differentiating flow regimes and make identification automatically with high accuracy.

2 | EXPERIMENT

2.1 | Experimental set-up

The experimental set-up is shown in Figure 1. The whole system consists of three specifically functioning parts, namely gas source part, underwater bubble flow generating part and signal acquisition, and storage part respectively. The working fluid used in the experiment is compressed air for gas and tap water for liquid.

The gas source part supplies the bubble flow generation part with compressed air of required thermal conditions. The air is compressed by the compressor and stored in a tank (0.8 m^3, 0.8 MPa) at a certain pressure (<0.8 MPa) in advance to ensure the stability of gas throughout the experiment. While experimenting, the air is let out from the tank and filtered to remove water and oil, then regulated by pressure regulating valve (SNS AR23000-03, China) and throttle to meet experimental requirements. The gas pressure in the experiment is set to be 0.05 MPa (Gauge pressure), and the flow rate varies within 3.45–-40.37 L min^{-1} (SIARGO MF5712, USA) to ensure the bubble flow experience all three flow regimes.

The bubble flow generation part is a water tank with an underwater exhaust at bottom. The dimension of the tank is 600 (L) × 600 (W) × 900 (H) mm^3, large enough to neglect the acoustic reverberation[29], which will be further verified by autocorrelation of the signal in the following chapter. The underwater exhaust is actually a plate with circular array of holes in the center, and some of the holes are blocked while the others are attached to tiny pipes so as to form a specific combination of gas outlet, which is, a different shape of bubble flow. The parameters of the pipes are listed in Table 1, and the layout of the pipes are shown in Figure 2a.

The signal acquisition and storage part includes hydrophones, high-speed camera, data acquisition, and analysis device and computer. The PAE signal emitted from the bubble flow is first sensed by three hydrophones (Reson TC4013, Denmark) fixed along the axial direction of the bubble flow, the layout of the hydrophones is depicted in detail in Figure 2b. The signals are recorded at a rate of 20,480 Hz for 20 s at each gas flow rate for five times. Then the signals travel to the data acquisition and analysis device (Econ AVANT MI-7016, China) and get converted from electrical signals to digital signals. Finally, the signals are transmitted and stored in a PC. At each flow rate, the video of bubble flow is shoot by a high-speed camera (NAC Memrecam HX-3) at a frame rate of 3,000 Hz (Resolution: 1280 × 960) simultaneously.

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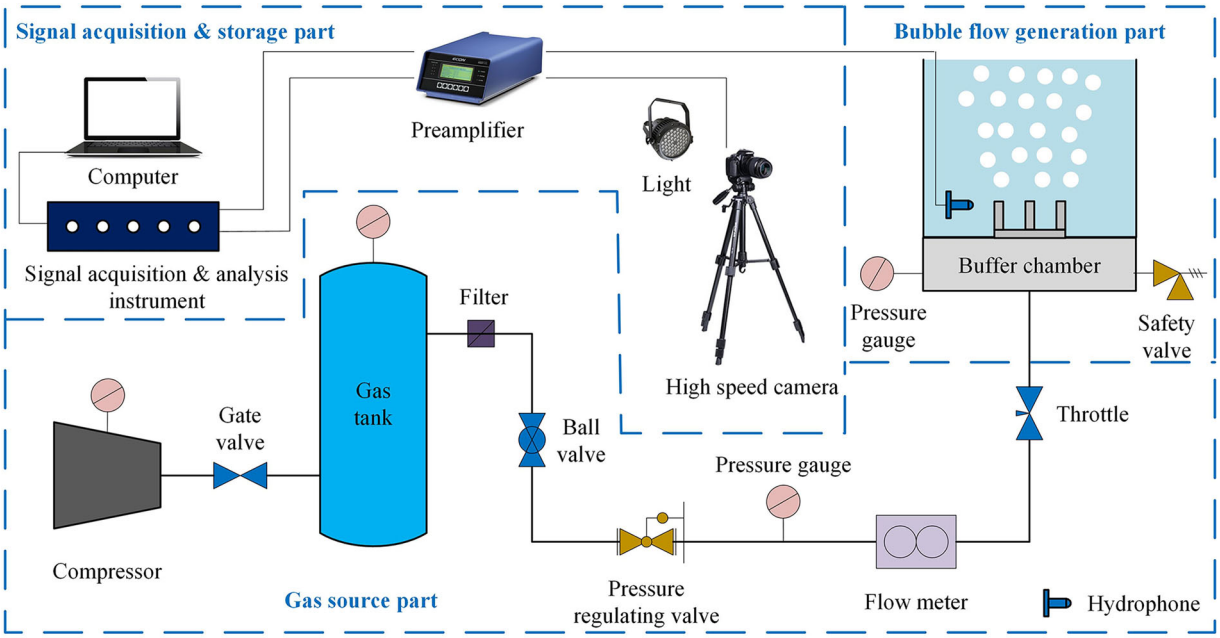


FIGURE 1 Experimental setup [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 1 Parameters of the pipes

Outer diameter (mm)	Inner diameter (mm)	No. of pipes	Array shape (line/circle)	Spacing (mm)
4.0	1.5	5	Line	10

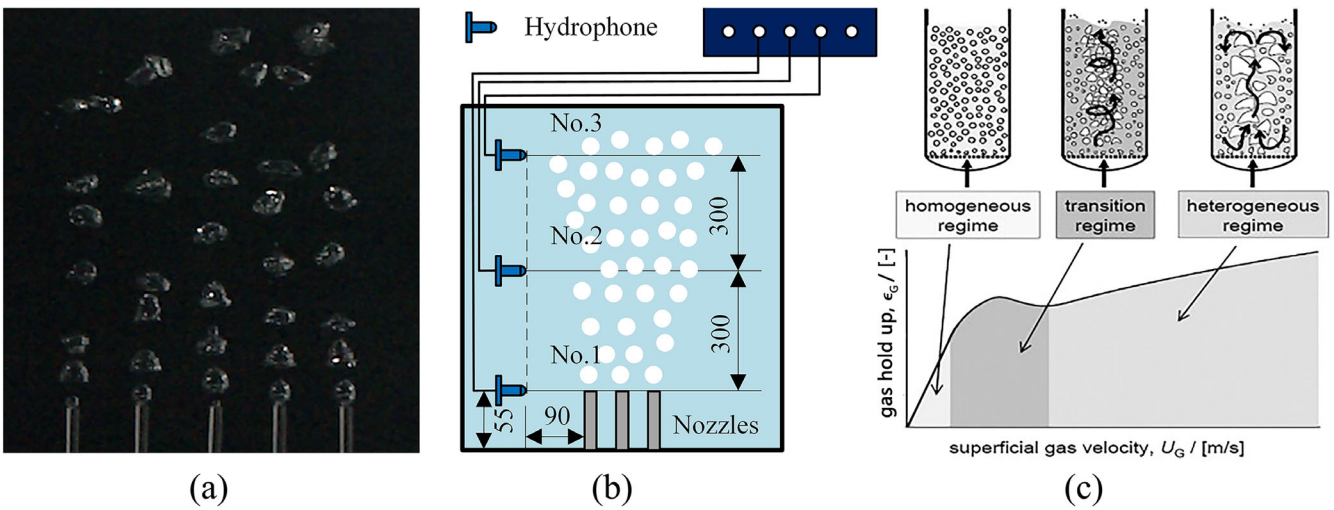


FIGURE 2 (a) Layout of the five pipes; (b) Layout of the hydrophones; (c) Commonly observed bubble flow regimes in bubble column[33] [Color figure can be viewed at wileyonlinelibrary.com]

2.2 | Bubble flow regime and regulation

In bubble columns commonly seen in process industries, there are usually three kinds of bubble flow regimes including homogeneous flow (bubbly flow), transition flow, and heterogeneous flow (churn turbulent),[30] as illustrated in Figure 2c. The homogeneous flow, generally known as bubbly flow, is characterized by uniformly spaced spherical bubbles of similar size in the liquid with practically no break-ups or coalescences. The heterogeneous flow, on the contrary, is an unstable flow regime with multitude bubbles of intense interactions, and break-ups and coalescences happening all the time, resulting in a wide range of bubble size distribution.[3] The transition flow is a regime observed between homogeneous and heterogeneous flow where the bubbles begin to coalesce and break up, not well defined as the other two.[31] Actually, there is one more flow

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regime called slug flow, where the bubbles exist in size nearly equivalent to the length scale of the container,[11] but it will not be discussed for not being observed in the experiment.

As can be seen from Figure 2c and known from existing researches, the bubble flow regime generally changes as a result of gas flow rate change. So, in the experiment, we obtained all three bubble flow regimes by changing air flow rate while keeping the air pressure and temperature constant. All three flow regimes observed in the experiment are shown in Figure 3, and the according condition parameters of the air in the experiment are listed in Table 2. The gas flow rates in the experiment are determined by the criterion that the resulting bubble flow regimes are typical enough for researchers to tell them apart just by the images recorded by high speed camera, which is the only way to pre-label the PAE signals accurately.

3 | METHODOLOGIES

Bubble acoustics, TF analysis and CNN are three main components of the bubble flow identification method proposed here. The bubble acoustics determines the relationship between bubble flow regime and the characteristics of its PAE signals, and is the premise for the establishment of the method. As for TF analysis, the SWT is chosen over STFT and continuous wavelet transform (CWT) for its superior TF resolution, and used to get characteristic-enhanced TFRs of sound signal to improve the identification accuracy. Moreover, the CNN works as a classifier and identifies different bubble flow regimes automatically with TFRs based datasets as input. The basic principles of bubble acoustics, SWT, CNN and the bubble flow regime identification method are explicitly explained in this section.

3.1 | Bubble acoustics and PAE signals

According to bubble acoustics, the detachment, breaking and coalesce behaviors of a bubble will excite its volume pulsation, thereby emitting sound signals into the surroundings, also known as the passive sound emission process of a bubble. The frequency of a bubble's PAE signal is called Minnaert frequency, described by Minnaert relation:

$$f_0 = \frac{1}{2\pi R_0}\sqrt{\frac{3\gamma P_0}{\rho}}, \tag{1}$$

where f_0 denotes volume pulsation frequency in Hz, R_0 is the bubble equivalent sphere radius, γ is the gas specific heat ratio, P_0 is the absolute liquid pressure, and ρ is the liquid density.[32] A bubble flow is consisted of bubbles with specific distribution, and its PAE signal undoubtedly comes from all the bubbles inside, but the frequency characteristics are different from a single bubble due to the acoustic interactions between bubbles. There are two major components in bubble flow's PAE signals based on the frequency spectrum, as depicted in Figure 4a: The higher one is Minnaert frequency excited by the detachment of bubbles from nozzles, and the lower one is the resonance frequency of the bubbles inside caused by bubble interactions.

The existence of Minnaert relation indicates a relatively rigor mathematical relationship between bubble flow regime and the frequency characteristics of its PAE signal. The bubble flow regime is a generalized description of the size and space distribution and interactions of the bubbles inside, and it is settled as long as the fluid and device parameters are settled. In other words, with given fluid parameters in a specific bubble column, the number, size and space distribution, and interactions of bubbles inside are determined. Under this circumstance, the Minnaert frequency of bubble flow is also determined according to the Minnaert relation, as is the resonance frequency, which is related to the Minnaert frequency through the number of bubbles in the bubble flow.[33] In all, based on bubble acoustics, the fluid parameters uniquely determine the bubble flow regime, and thus uniquely determine the frequency characteristics of the bubble flow PAE signals in a given bubble column, and vice versa, so it is

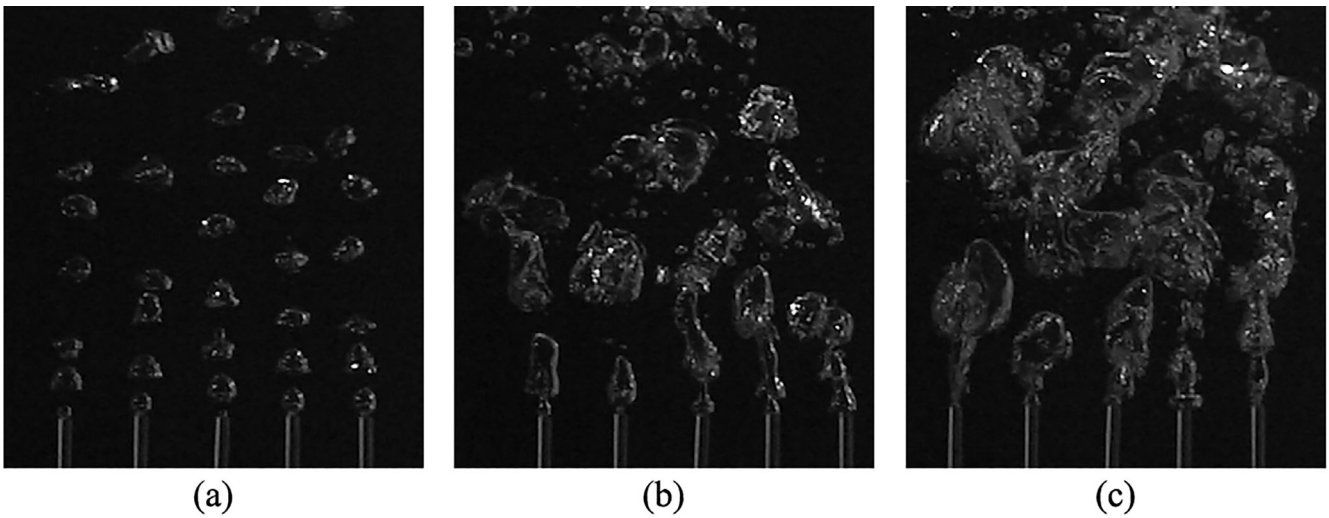


FIGURE 3 Three bubble flow regimes observed in the experiment: (a) Homogeneous flow; (b)Transition flow; (c) Heterogeneous flow

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TABLE 2 Parameters of the compressed air

No.	Temperature (K)	Pressure (MPa)	Flow regime	Flow rate (L min^{-1})	Gas velocity (m/s)	Reynolds number
1	298.15	0.05	Homogeneous	1.83	3.45	490.44
2			Transition	8.02	15.13	2,150.84
3			Heterogeneous	21.40	40.37	5,738.90

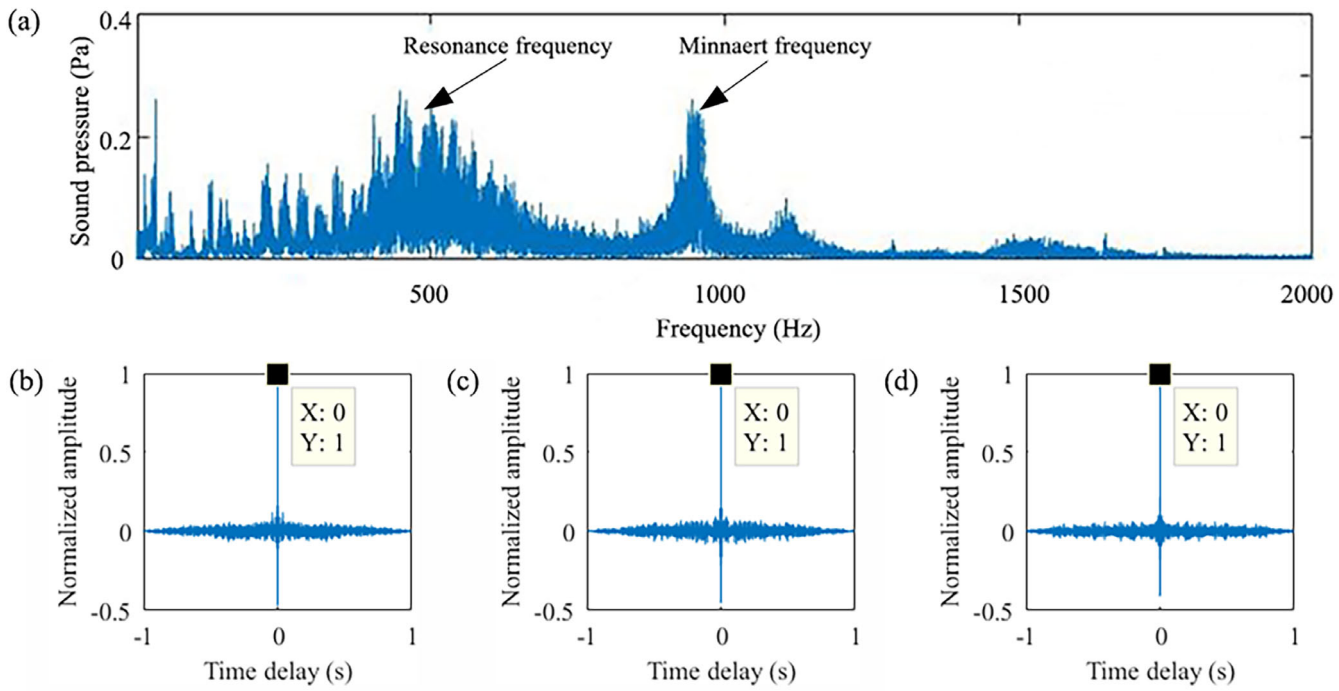


FIGURE 4 (a) Frequency spectrum of bubble flow; (b) Autocorrelation function of homogeneous flow's passive acoustic emission (PAE) signal; (c) Autocorrelation function of transition flow's PAE signal; (d) Autocorrelation function of heterogenous flow's PAE signal [Color figure can be viewed at wileyonlinelibrary.com]

reliable to use the TFR of bubble flow PAE signal to identify the bubble flow regime in a bubble column.

For the PAE signal to identify bubble flow regimes well, it has to be strong enough compared with the background noise, that is, have a sufficiently high signal to noise ratio (SNR). The higher the SNR of the PAE signal, the more obvious the key acoustic characteristics of the signal, and the more accurate the result of bubble flow regime identification based on the signal. The threshold of SNR in practical applications depends on the acquisition cost and identification accuracy requirements. During our experiment, three PAE signals are acquired by three hydrophones fixed close enough to the bubble flow without affecting the flow field synchronously, and it is found that they are from one same source by analyzing and contrasting, of which the signal from No. 1 hydrophone (see in Figure 2b) is the strongest, so all the analysis in the paper is based on this signal. Before being used to identify different flow regimes, the sound signals are first autocorrelated to examine the effects of reverberation. The hydroacoustic reverberation is caused by the scattering and reflection of sound signals underwater, and are generally superposition of signals similar to direct sound but with different time delays in time domain. The autocorrelation operation, meanwhile, are usually used to evaluate the similarity between signal and its time-domain translation by convolution, thus the perfect choice for detecting reverberation in the signal. The autocorrelation results of the measured signals are shown in Figure 4b–d. For the sound signals of all three regimes, the maximum value is obtained at zero and then quickly attenuates in the vicinity, which shows that the main component of the signal is direct sound, and the intensity of the reverberation component is very weak relatively and can be ignored.

3.2 | Synchrosqueezing wavelet transform

The SWT is a TF signal analysis method based on WT and reallocation. First proposed by Daubechines et al.,[28] SWT is derived to solve the TF blurring phenomenon in WT by squeezing and redistributing the wavelet coefficients acquired by WT.

Given a signal $s(t)$, the continuous WT $W_s(a, b)$ of $s(t)$ is defined as

$$W_s(a,b) = \int s(t) a^{-1/2} \psi\left(\frac{t-b}{a}\right) dt, \quad (2)$$

where a is scaling factor; b is time offset factor; $\psi(t)$ is a properly chosen wavelet basis; $\bar{\psi}(t)$ is the complex conjugate of $\psi(t)$.

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If $s(t)$ is a purely harmonic, let us say $s(t) = A\cos(\omega t)$, and take $\hat{\psi}(\xi)$, the Fourier transform of $\psi(t)$ to satisfy $\hat{\psi}(\xi) = 0$ when $\xi < 0$, then $W_s(a, b)$ can be written as

$$\begin{aligned} W_s(a,b) &= \frac{1}{2\pi}\int \hat{s}(\xi)a^{1/2}\hat{\psi}(\bar{a}\xi)e^{ib\xi}d\xi \\ &= \frac{A}{4\pi}\int[\delta(\xi-\omega)+\delta(\xi+\omega)]a^{1/2}\hat{\psi}(\bar{a}\xi)e^{ib\xi}d\xi \\ &= \frac{A}{4\pi}a^{1/2}\hat{\psi}(\bar{a}\omega)e^{ib\omega}, \end{aligned} \tag{3}$$

where $\hat{s}(\xi)$ is the Fourier transform of $s(t)$, and $\hat{\psi}(\bar{a}\xi)$ the complex conjugate of $\hat{\psi}(a\xi)$, and $\delta(\bullet)$ the delta function. With reference to Equation (3), a candidate instantaneous frequency $\omega_s(a, b)$ of signal $s(t)$, if $W_s(a, b) \neq 0$ at any point (a, b), can be calculated by

$$\omega_s(a,b) = -i(W_s(a,b))^{-1}\frac{\partial}{\partial b}W_s(a,b), \tag{4}$$

this way, the scale-time plane (a, b) is transformed into frequency-time plane $(\omega_s(a, b), b)$. Since the WT $W_s(a, b)$ is usually calculated only at discrete values a_k, the synchrosqueezing transform $S_s(\omega_l, b)$ at any center frequency ω_l is obtained by squeezing the values in nearby interval $\left[\omega_l - \frac{1}{2}\Delta\omega, \omega_l + \frac{1}{2}\Delta\omega\right]$ to ω_l at a_k

$$S_s(\omega_l,b) = (\Delta\omega)^{-1}\sum_{a_k:|\omega(a_k,b)-\omega_l|\le\frac{\Delta\omega}{2}} W_s(a_k,b)a_k^{-3/2}(\Delta a)_k, \tag{5}$$

where $(\Delta a)_k = a_k - a_{k-1}$; $\Delta\omega = \omega_l - \omega_{l-1}$. The blurring region in WT is squeezed into a concentrated line and the TF resolution hence gets highly improved. The SWT is also reversible like fast Fourier transform (FFT), and the readers may refer to Daubechines's[28] article for more details.

The time series and TFRs of the sound signals of homogeneous flow are shown in Figure 5, where picture (a)–(d) are time series, TFR by STFT, TFR by CWT, and TFR by SWT respectively. As can be seen, all three TFRs have the common feature of periodic vertical linear spots, and the resolution difference among three TFRs shows the great advantage SWT has over STFT and CWT on revealing the TF details of the signal.

And in Figure 6, the TFRs by SWT of homogeneous flow, transition flow and heterogeneous flow are given. It is easy to observe from the pictures that different bubble flow regimes have distinctive TFR textures, and the textures change from periodic vertical linear spot to periodic left parabolic spot as the flow regime transforms, thus it is reasonable to use TFRs based datasets of sound signal as the input of CNN to identify bubble flow regimes.

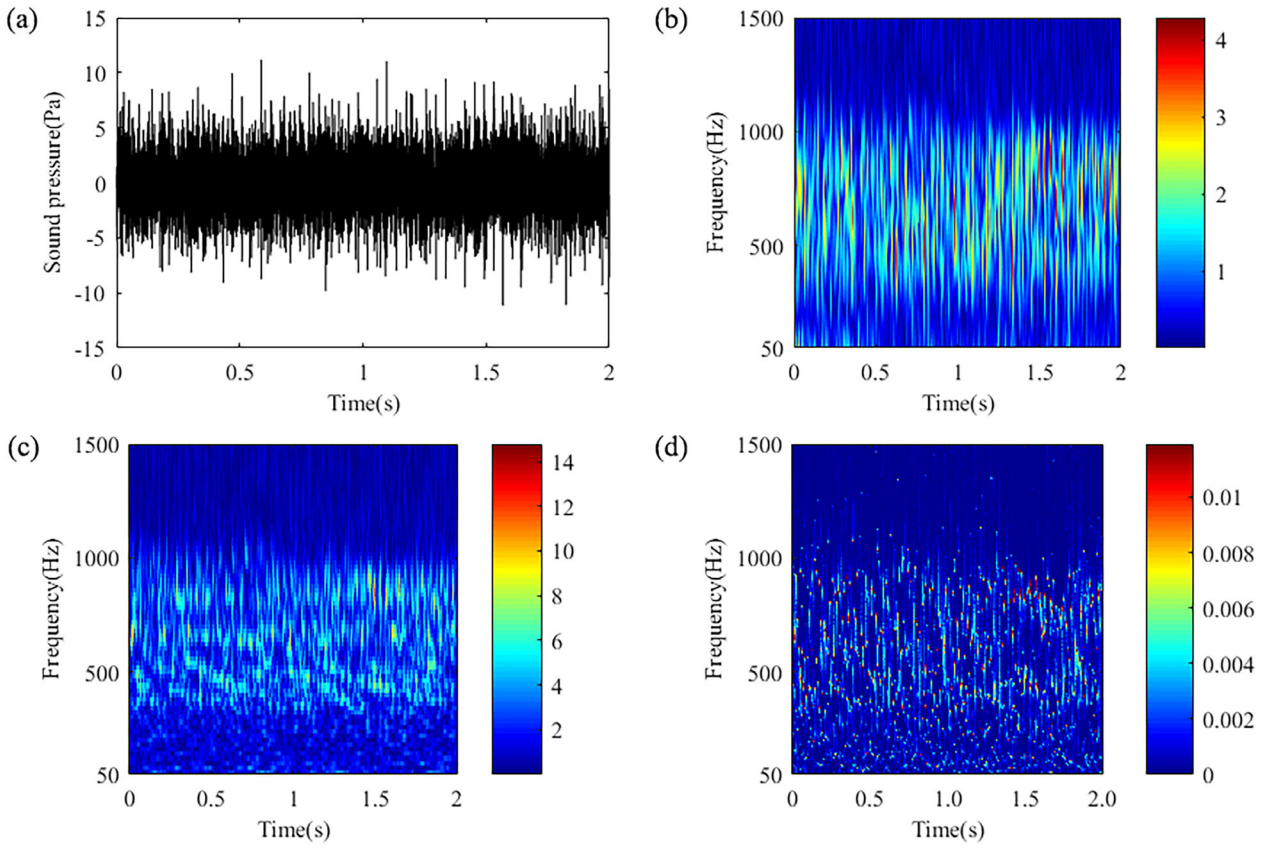


FIGURE 5 Time series and time-frequency representations (TFRs) of the sound signal of homogeneous flow: (a) Time series; (b) TFR by short time Fourier transform; (c) TFR by CWT; (d) TFR by synchrosqueezing wavelet transform [Color figure can be viewed at wileyonlinelibrary.com]

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3.3 | Convolutional neural network

The CNN is a DL architecture commonly used for image recognition and classification.[21,22,24,34] It is essentially a multilayer neural network with convolutional structures, highly invariant to translation, scaling, tilting, or any other kinds of deformation of two-dimensional images. A typical CNN has the architecture consisting of an input layer, alternating convolutional and pooling layers, fully connected layers and an output layer from bottom to top.

Convolutional layers extract features from input map with multiple learnable convolution kernels.[35] While working, the input data is "convolved" by the kernels and then processed by an activation function to get output feature maps.[34] The convolution between input map and the kernel is actually cross-correlation mathematically, one kind of kernel operation, and is illustrated intuitively in Figure 7a: Suppose that the input is a 4×4 grayscale image, and the kernel is a 3×3 matrix. Starting from the upper left corner of the input image, calculate the sum of element-wise products of the filter and the first 3×3 block of the image, and the first element of the feature map is obtained, then the rest elements can be obtained by moving the block rightward or downward with a stride of one and repeating the kernel operation.

After each kernel operation, it is easy to see that not only output gets smaller, but also the information on the edges is discarded, and if the size of the input or the information edges contain need to be preserved, then the zero padding shown in Figure 7b is introduced. It is an operation where borders of pixels all with value zero are added around the edges of input images. It enlarges the dimension of input images according to the dimension of the kernel thus allows us to preserve the size of input as well as extract information on the edges.

As can be seen above, the convolution operation is a linear transformation, while the reality is that most classification tasks are nonlinear, so the activation function is used to introduce nonlinearity into the network to improve its classification ability. The commonly used activation functions are sigmoid function, tanh function and ReLU function, depicted in Figure 7c, and the ReLU function is our final choice for its significant advantages. The output feature map of the lth convolutional layer x_j^l can be described as[27]

$$x_j^l = f\left(\sum_{i=1}^{M} x_j^{l-1} * k_{ij}^l + c_j^l\right), j = 1, 2, \ldots, N, \tag{6}$$

where $f(\bullet)$ denotes the activation function; the superscript l is the ordinal number of the layers, and the subscript i and j are the ordinal number of the input feature map and convolution kernel respectively; x_j^{l-1} is the input map of the lth layer; k_{ij}^l is the jth convolution kernel connected to the ith input map; c_j^l is the offset coefficient corresponding to jth kernel; * denotes convolution operation.

The pooling layer, following the convolutional layer, also consists of multiple feature maps, each of which uniquely corresponds to the feature map of upper layer, thus the number of the feature map is not changed. Pooling is a subsampling operation, and is accomplished by dividing the image into many blocks of the same size (pooling size) without overlapping, and representing each block's feature by its statistics. The image after pooling is much smaller than the original image while retaining the original information. The commonly used pooling methods are maximum pooling and average pooling, which represent each block's feature with its maximum and average respectively.[36,37] The pooling layer aims at obtaining spatial invariant features by reducing the dimension of feature map.[38] The whole pooling process is illustrated in Figure 8a, and can be written as[27]

$$x_j^l = f\left(\beta_j^l down\left(x_j^{l-1}\right) + c_j^l\right), j = 1, 2, \ldots, M, \tag{7}$$

where $f(\bullet)$ denotes the activation function; $down(\bullet)$ denotes the down-sampling function; β_j^l is the weight coefficient of the pooling layer.

The last layer of CNN is always a fully connected layer together with a softmax classifier. Each neuron in this layer is fully connected to all neurons in the upper layer, shown in Figure 8b, just like in the multilayer perceptron, and the softmax normalizes the outputs of the fully connected layer and outputs the probability of each type.[21]

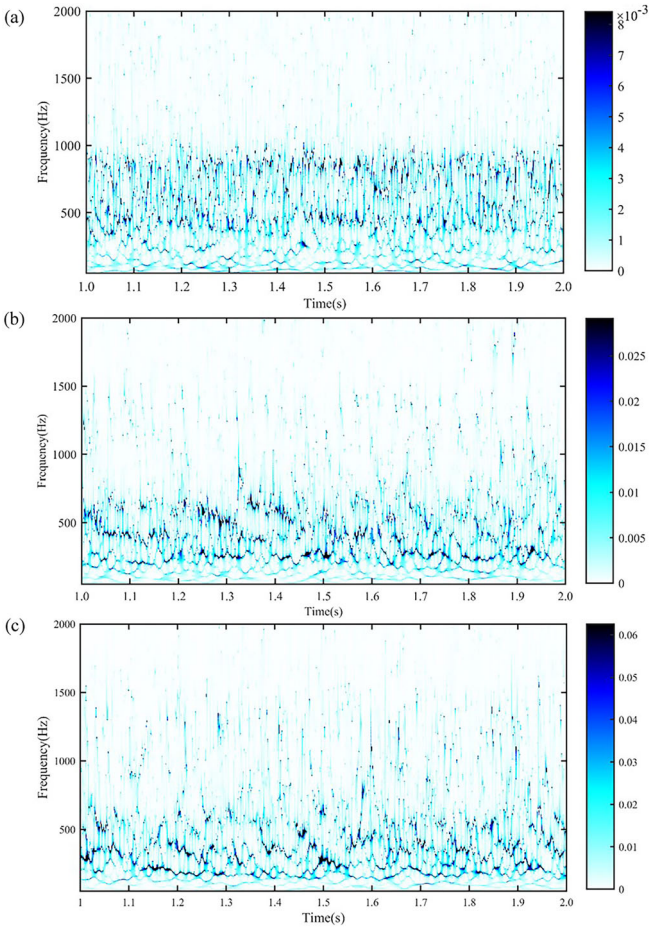


FIGURE 6 Time series and time-frequency representations of different bubble flow regimes: (a) Homogeneous flow; (b) Transition flow; (c) Heterogeneous flow (Shallow background) [Color figure can be viewed at wileyonlinelibrary.com]

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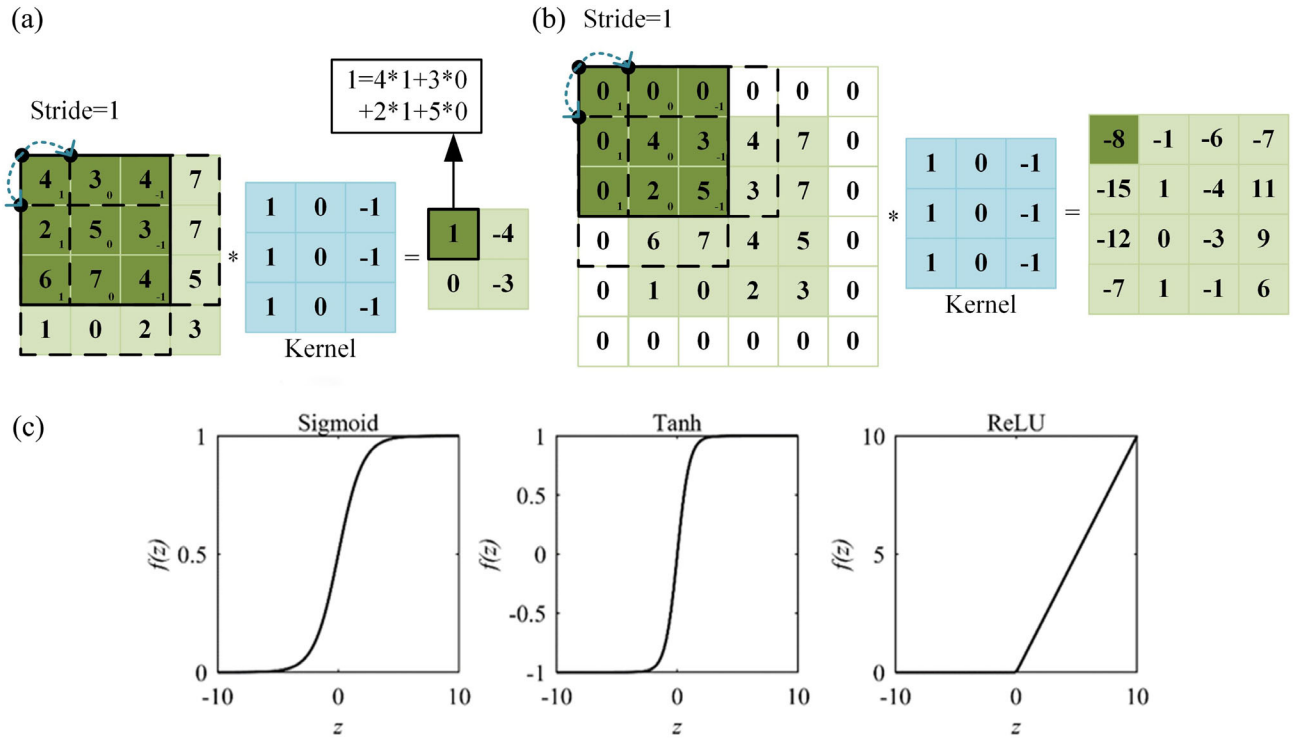


FIGURE 7 (a) Convolution operation; (b) Convolution operation with zero-padding; (c) Commonly used activation functions [Color figure can be viewed at wileyonlinelibrary.com]

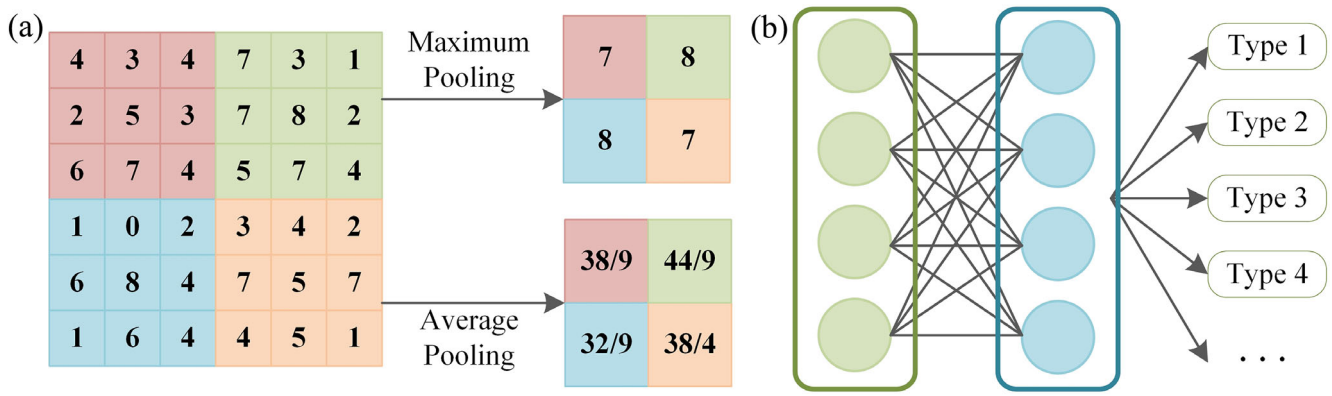


FIGURE 8 (a) Maximum and average Pooling operation; (b) Fully connected layer [Color figure can be viewed at wileyonlinelibrary.com]

The input of the fully connected layer is a one-dimensional vector by flatting the learned feature maps,[39] and the output x_j^l can be obtained by[27]

$$x_j^l = f\left(\sum_{i=1}^{M} x_j^{l-1} \times w_{ij}^l + c_j^l\right), j = 1,2,\ldots,N, \tag{8}$$

where $f(\bullet)$ denotes the activation function; w_{ij}^l is the weight coefficient of the jth output value connected to the ith input value.

The overall architecture of a CNN is depicted in Figure 9. The CNN architectures used are obtained by fine-tuning three typical CNN architectures, namely AlexNet, VggNet, and ResNet (so-called transfer learning): All parameters in the CNN architectures are frozen except the last layer of the fully connected layer, which is tuned to meet the needs of three bubble flow regimes identification. The architecture of AlexNet, VggNet, and ResNet are shown in Figure 10, and the modifications made on the fully connected layer together with the number of parameters that need to be trained after fine-tuning are given in Figure 11.

3.4 | Bubble flow regime identification method combining SWT and CNN

The classification and definition of different bubble flow regimes is the premise of all identification methods, but it also tricks most

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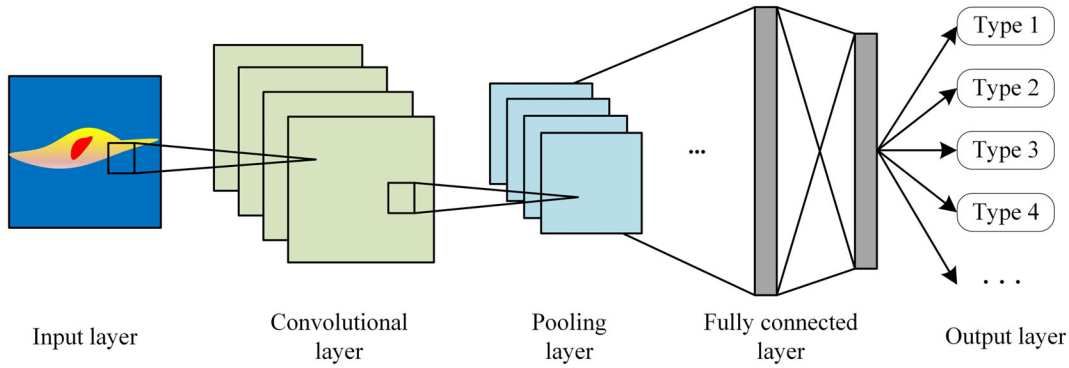


FIGURE 9 Overall structure of a convolutional neural network [Color figure can be viewed at wileyonlinelibrary.com]

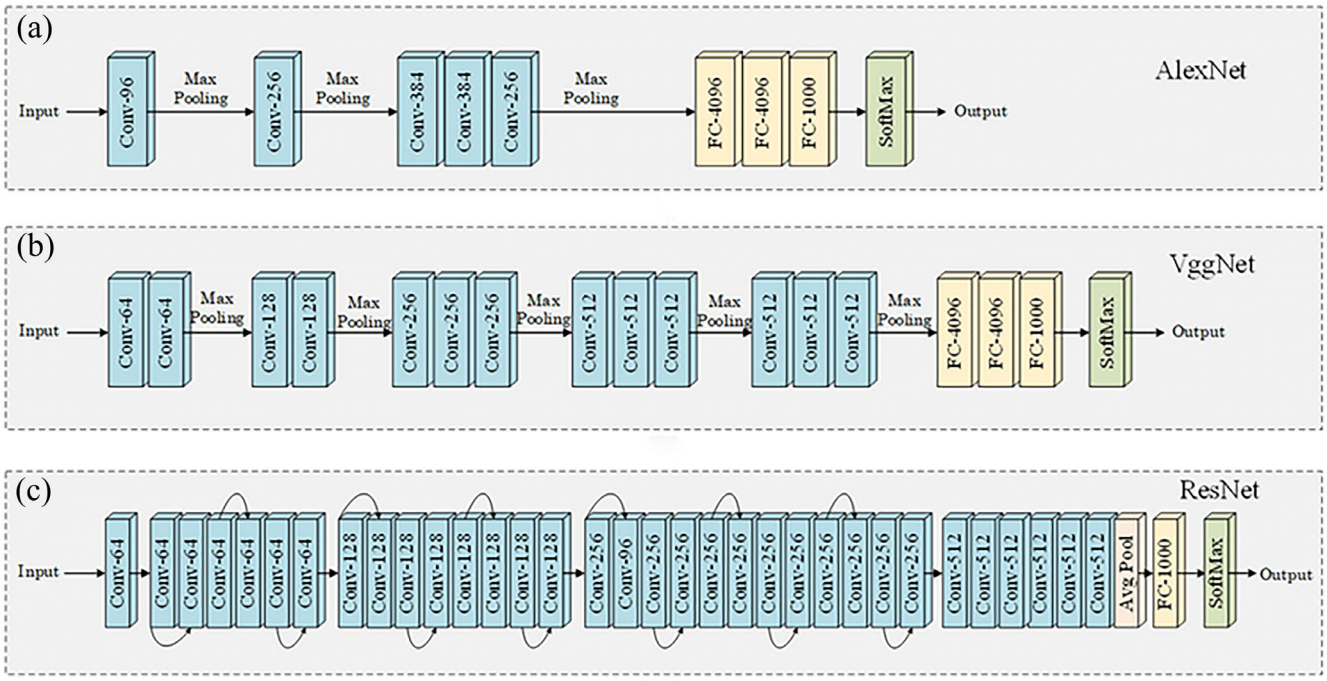


FIGURE 10 Three typical convolutional neural network architectures: (a) AlexNet; (b) VggNet; (c)ResNet [Color figure can be viewed at wileyonlinelibrary.com]

identification methods into the bottleneck of finding the exact boundaries between different flow regimes by concealing the fact that the bubble flow change is continuous. The bubble flow identification method combining SWT and CNN avoids the bottleneck by focusing on the key acoustic characteristics differentiating different bubble flow regimes. It is established on the relationship between bubble flow regime and the acoustic characteristics of its PAE signal, and extracted the key acoustic characteristics from the PAE signals of typical bubble flow regimes with SWT and CNN, and tell us the probability of a bubble flow belonging to each bubble flow regime by contrasting the key characteristics instead of determining the boundaries. The specific steps of the method can be described as follow: (a) Measure and store the PAE signals of each bubble flow; (b) Filter the signals and obtain TFRs through SWT; (c) Slice the TFRs to thousands of segments of specific scale so as to build a dataset consisting of a training set, a validation set and a test set; (d) Label the dataset with bubble flow regime by images recorded in high-speed camera; (e) Train, validate and test the CNNs with dataset to extract the key TF textures of each flow regime and get optimal results. One point about the method that needs special explanation is that the CNNs are tuned with data from typical states (images highly identifiable by human eyes) of three bubble flow regimes so as to ensure the accurate label of the data and exact extraction of the key acoustic characteristics. The flowchart of the whole process is depicted in Figure 12.

The parameters of the dataset depend on the parameters and textures of the TFRs acquired. As mentioned above, the sound signals are recorded at a rate of 20,480 Hz for 20 s at each gas flow rate for five times, so the full size of each TFR is 20 s × 10,240 Hz. But for the TF textures of concern, the frequency is in the range of [0, 2000] Hz, the sound signal of bubble flow in the experiment, and the time scale

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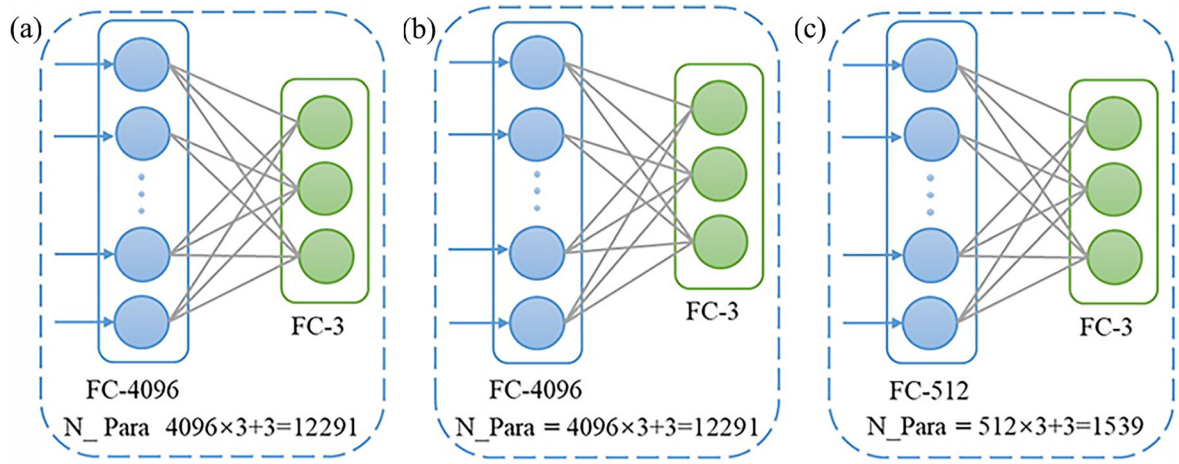


FIGURE 11 Modification made on the fully connected layer: (a) AlexNet; (b) VggNet; (c) ResNet, and N_Para stands for the number of parameters need to be trained [Color figure can be viewed at wileyonlinelibrary.com]

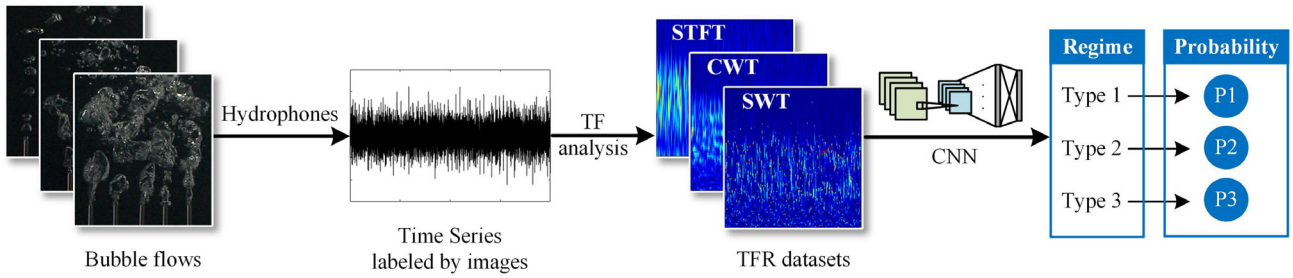


FIGURE 12 Flowchart of passive acoustic emission, synchrosqueezing wavelet transform, and convolutional neural network based bubble flow regime identification [Color figure can be viewed at wileyonlinelibrary.com]

TABLE 3 Parameters of the TF texture dataset

	Homogeneous flow	Transition flow	Heterogeneous flow
Dataset element	0.5 s (time) × 2000 Hz (frequency) slice of TFR		
Size of dataset	1,000	1,000	1,000
Size of training set	700	700	700
Size of validation set	200	200	200
Size of test set	100	100	100

Abbreviations: TF, time-frequency; TFR, TF representation.

TABLE 4 Final hyperparameters of three CNNs based on SWT TF texture dataset

CNN model	Batch size	Epoch	Learning rate	Optimization algorithm	Loss function
AlexNet	5	100	0.00005	Adam	Cross entropy loss function
VggNet			0.0001		
ResNet			0.00005		

Abbreviations: CNN, convolutional neural network; TF, time-frequency; SWT, synchrosqueezing wavelet transform.

is no longer than 0.5 s, so the scale of TFRs slices is determined to be 0.5 s × 2000 Hz. And the dataset is built by slicing TFRs into thousands of segments of above size. The parameters of the dataset are listed in Table 3. As can be seen, the entire dataset mainly includes three parts: the training set, the validation set, and the test set, corresponding to the three steps of neural network optimization: training, validation, and testing.

4 | RESULTS AND DISCUSSIONS

The CNNs in this paper are constructed and trained with Pytorch DL framework. During each iteration, only a small part of the training set, that is, a batch, passes through the network to complete one forward propagation and one backward propagation to update the weights in CNN. Every time the batches cover all the data in the training set, an

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epoch forms. For each epoch, the difference between true values and predicted values is evaluated by loss function as Loss. By observing how Loss and Accuracy change with the number of epochs during the training process, the hyperparameters and optimization algorithm can be adjusted. The same holds true for the validating process on the validation set. The final values of hyperparameters of three fine-tuned CNNs after being trained are shown in Table 4.

Bias and variance (VAR) are two important parameters for evaluating the fitting state of the neural network, where the bias is the difference between model output and the true value and VAR the difference of output between different models. The combination of small bias and small VAR indicates a well-fitted model. In the method proposed, the difference between training error and validation error can be seen as the VAR, and the VAR satisfies

$$\text{error} = \text{bias}^2 + \text{VAR}, \tag{9}$$

where error refers to training error. In Figure 13, curves showing how training loss, training accuracy, validation loss and validation accuracy change as the number of epochs increases for AlexNet, VggNet, and ResNet trained with SWT TF texture datasets are given. As can be seen from Figure 13a–c, each CNN architecture shows good convergence, small bias, and VAR, implying good fit for all three. But among all CNNs, the AlexNet convergences the fastest, and the ResNet holds the highest validation accuracy of 100% from Figure 13d.

After training and validation, the performance of the three trained models is finally tested on the test set and the result is shown as confusion matrixes in Figure 14. The identification accuracy of AlexNet, VggNet, and ResNet is 97.66%, 98.00%, and 99.33%, respectively, and the ResNet shows the strongest identification capability. In daily application scenarios, the construction, and optimization of the method are usually all set before actual usage, thus the identification capability of the method matters more than the convergence speed, so the ResNet performs the best in bubble flow regime identification.

To further study the influence of TF analysis method on the identification accuracy, the TFRs obtained by STFT and CWT are also used to tune those three CNN models, and the final identification accuracy is depicted in Figure 15a. Four conclusions can be drawn from the picture: (a) The TF analysis method has stronger impact on the identification accuracy than the CNN models; (b) The SWT always gives the

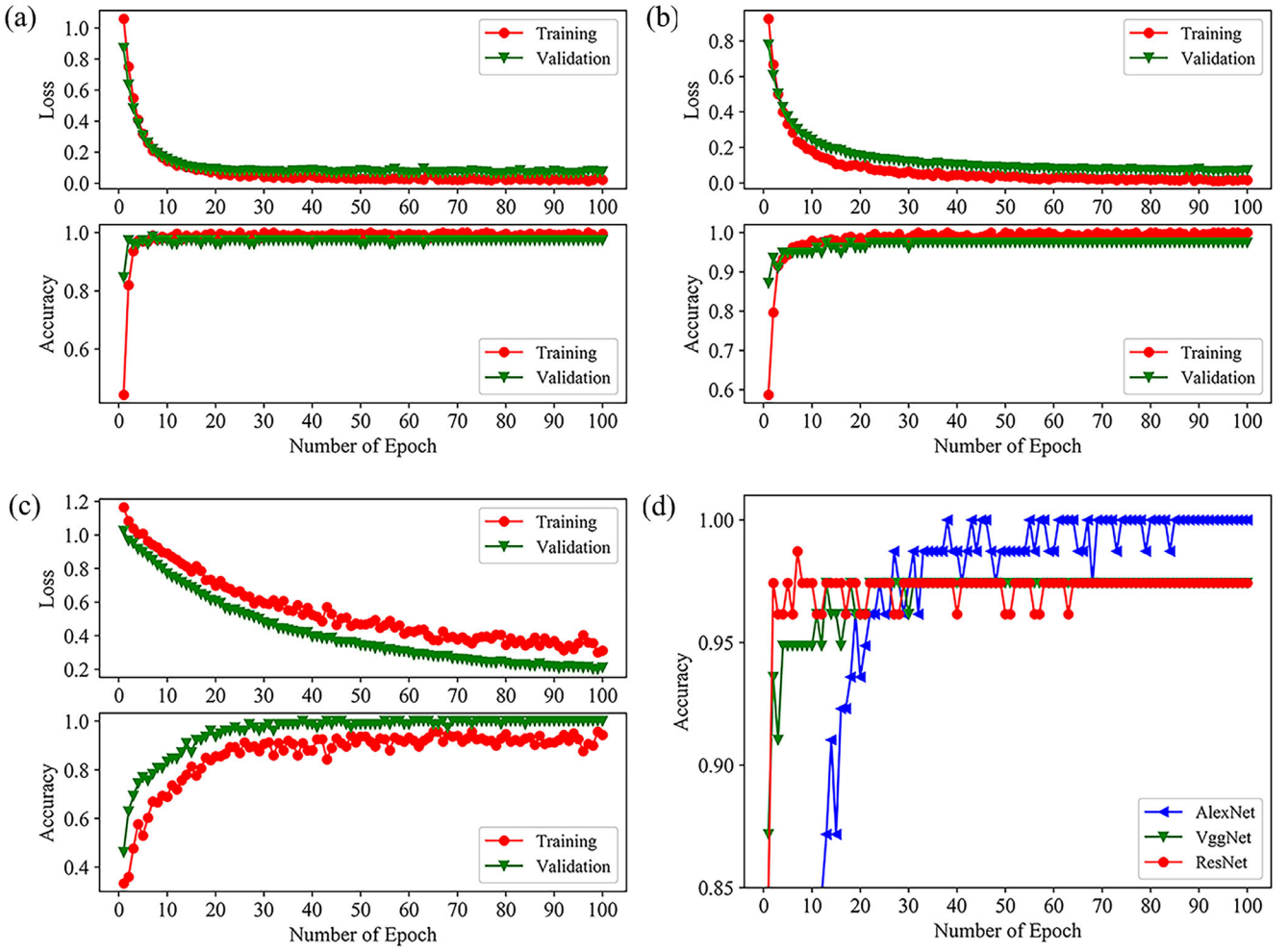


FIGURE 13 Training and validation process of convolutional neural networks based on synchrosqueezing wavelet transform dataset: (a) AlexNet; (b) VggNet; (c) ResNet; (d) Validation accuracy of AlexNet, VggNet, and ResNet [Color figure can be viewed at wileyonlinelibrary.com]

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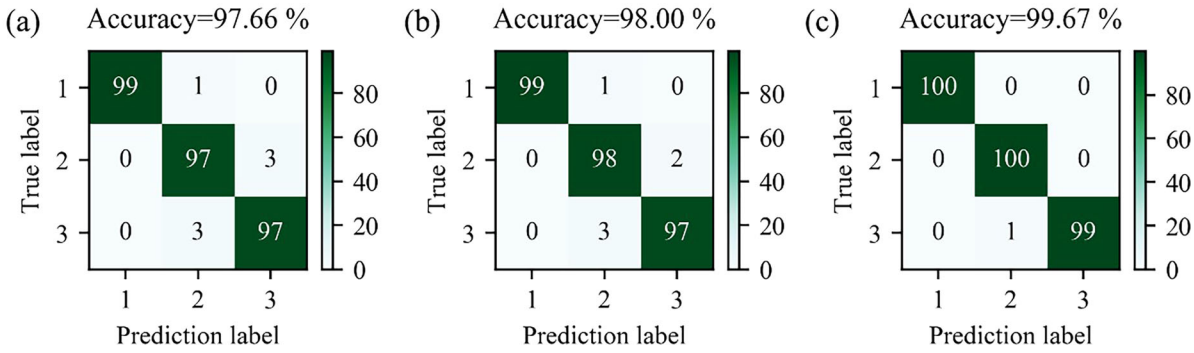


FIGURE 14 Confusion matrixes of (a) AlexNet; (b) VggNet; (c) ResNet; Label 1, 2, and 3 are for homogeneous, transition, and heterogeneous flow [Color figure can be viewed at wileyonlinelibrary.com]

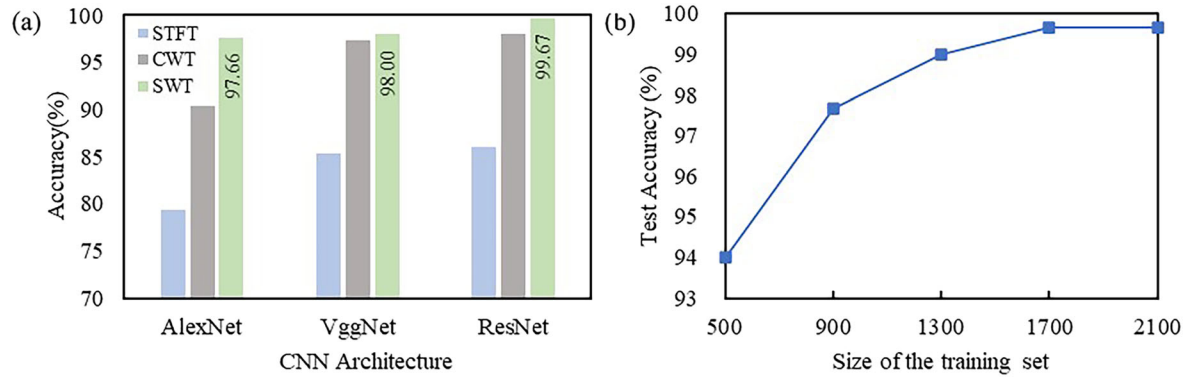


FIGURE 15 (a) Test accuracy of AlexNet, VggNet, and ResNet; (b) Test accuracy versus size of the training set (synchrosqueezing wavelet transform-ResNet method) [Color figure can be viewed at wileyonlinelibrary.com]

best identification accuracy in three TF analysis methods; (c) ResNet performs better than the other two CNN models; (d) The combination of SWT and ResNet contributes the highest identification accuracy. Overall speaking, the PAE based method combing SWT and ResNet (PAE-SWT-ResNet) serves as the best choice for bubble flow regime identification.

In addition, the reasonability of the training set size for PAE-SWT-ResNet method is also evaluated by comparing the test accuracy under different set sizes. The result is shown in Figure 15b, and it is obvious that the test accuracy increases rapidly at first and stabilizes near the upper limit after the set size reaches a certain value of 1,700, so it is safe to say that the results obtained from training set sizing 2,100 is fully reliable.

5 | CONCLUSION

In this paper, a passive acoustic method based on SWT and CNN is proposed to automatically identify bubble flow regime in bubble columns with high identification accuracy. The proposed method is established on the fact that the bubble flow radiates PAE signal when generated, and the TF texture of the PAE signal change with bubble flow regime, just like how fingerprint differs among different persons.

For the method to work, the PAE signals of bubble flow are first acquired by three hydrophones mounted in a controllable underwater experimental set-up, and then converted into TFRs through TF analysis, from which TF textures datasets can be built. Next, the TF textures datasets are labeled with bubble flow regime by the images recorded in the high-speed camera. Finally, the CNNs are constructed by fine-tuning typical CNN architectures, and then trained, validated and tested by the datasets to extract the key acoustic characteristics of bubble flow and identify the bubble flow regime automatically.

The effects of the TF analysis method and CNN architecture on the identification accuracy are studied. By redistributing and squeezing the wavelet coefficients obtained by WT to minimize the blurring in TFRs, the SWT outstands traditional TF analysis methods such as STFT and CWT, and obtains TFRs with the best TF resolution and clearest textures. Of all three fine-tuned typical CNN architectures, the ResNet performs the best no matter combined with which TF analysis methods. And the combination of SWT and ResNet contributes the highest identification accuracy and the most robust performance compared with other combinations of TF analysis methods and CNN architectures with no surprise.

The method can be easily translated to industrial bubble columns just by replacing the experimental data with production data and

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repeating the method with modifications of the SWT and ResNet parameters, and the whole process will not take long. The thick metal wall of the real bubble column may seem to be a big problem for acquiring sound signals, but it can be solved by holes on the wall or corrosion protection of the hydrophones, and so forth. One more thing to mention is that the TFR is different even for the same flow regime in two columns of different sizes due to the bubble number difference, which means the relationships between the bubble flow regime and TFR of those two columns are different. So, a trained CNN can only be shared among bubble columns having the same or similar parameters, fluid, and operating range with no need to be retrained, which is easy to meet in modern industry.

Overall speaking, the bubble flow regime identification method established in this paper is a quite new attempt. Compared with existing identification methods, this one combing PAE, SWT, and ResNet is much simpler and more economical, and most importantly, capable of identifying bubble flow regime automatically with an accuracy up to 99.67% and almost no subjective interference. With some adaptations to actual scenarios and quantitative explorations on strength threshold of the PAE signal, it is foreseeable that this method can function well in process industries as an online monitoring tool for bubble flow regimes in the near future.

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AUTHOR CONTRIBUTIONS

Ling Zhang: Conceptualization; data curation; formal analysis; investigation; methodology; resources; software; validation; visualization; writing-original draft; writing-review and editing. **Yun Xing:** Conceptualization; data curation; formal analysis; investigation; methodology; resources; software; validation; visualization. **Dazhuan Wu:** Conceptualization; data curation; formal analysis; funding acquisition; investigation; methodology; project administration; resources; supervision; writing-original draft; writing-review and editing. **Ning Chu:** Formal analysis; funding acquisition; investigation; project administration; resources; supervision.

DATA AVAILABILITY STATEMENT

Data available on request from the authors

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Ling Zhang https://orcid.org/0000-0002-6467-7067

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